

An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific REID, SEP acute hazard, and dual-use proton therapy RBE benchmarking

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Abstract

We describe GCR-DOSIMETRY-PIPELINE, an open-source Python package for estimating galactic cosmic ray (GCR) radiation exposure and cancer risk on crewed deep-space trajectories. The pipeline uses Gleeson–Axford force-field solar modulation [1] with per-species local interstellar spectra [2, 3]. GCR flux is transported through slab shielding via continuous-slowing-down approximation (CSDA) with effective-charge stopping power scaling; secondary neutron dose equivalent comes from a pre-computed HZETRN table [4]. Organ-specific cancer risk is estimated as radiation exposure in incidence of death (REID) across eleven organs at ICRP-89 tissue depths [5].

The pipeline’s per-species normalization factors are calibrated to the MSL/RAD cruise-phase measurement of 1.84 ± 0.33 mGy/day [6] at 16 g cm^{-2} aluminum, and the shielding-depth dependence and material ratio are independently compared against HZETRN benchmark data [4, 7], showing agreement within the stated benchmark

uncertainties ($\lesssim 20\%$ on absolute dose). A Solar Energetic Particle module estimates BFO absorbed dose from three canonical events using Band-function spectra [8]; the August 1972 event requires $> 30 \text{ g cm}^{-2}$ aluminum to fall below the NASA 30-day limit of 250 mGy.

Global physics and radiobiological uncertainties are propagated via Latin Hypercube Sampling over eight parameters, yielding a median REID of 5.0% with p5–p95 bounds of 1.3%–16.9% ($13.4\times$ spread, typical for this class of analysis) for a 35-year-old male crew member on a 259-day Earth–Mars transit at 16 g cm^{-2} aluminum. Variance decomposition shows that the quality factor scale parameter (76% of REID variance) and DDREF (7%) together dominate output uncertainty, while all GCR physics parameters together contribute $< 9\%$, confirming that better transport physics alone will not narrow REID bounds without matching progress in high-LET human radiobiology.

Per-species $\bar{L}_D$ decomposition reveals that carbon ions contribute approximately 59% of unshielded GCR absorbed dose at a field $\bar{L}_D \approx 18.7 \text{ keV } \mu\text{m}^{-1}$, yielding $Q_{\text{eff}} = 2.60$, consistent with the MSL/RAD measured value of 2.62 ± 0.14 within measurement uncertainty. Sixteen g/cm^2 of aluminum reduces the carbon fraction to 31% and the field $\bar{L}_D$ to $4.1 \text{ keV } \mu\text{m}^{-1}$, bridging toward the LET range of clinical proton therapy Bragg peaks. Wedenberg [9] and McNamara [10] proton RBE models agree with OpenTOPAS-RBE scorer output to within mean absolute error $< 10^{-3}$ for V79 cells (self-consistency check for clinical cell lines), enabling consistent comparison of space and clinical radiation quality factors from a single codebase. All source code, 103 automated tests, validation data, and TOPAS geometry files are freely available at <https://github.com/aryanhshah8/gcr-dosimetry-pipeline>.

Keywords: galactic cosmic rays; Mars mission dosimetry; radiation risk; REID; solar energetic particles; proton therapy RBE; uncertainty quantification; open-source

1 Introduction

Deep-space human exploration beyond low-Earth orbit exposes crew members to galactic cosmic rays (GCR) and solar energetic particles (SEP) at fluence rates and energies far exceeding the shielding capacity of current crew vehicles. The GCR environment at 1–1.5 AU

consists of protons, helium, and heavier ions (the so-called HZE particles) spanning kinetic energies from a few hundred MeV/n to tens of GeV/n, with the peak in biological effectiveness occurring near the energy range where stopping power is highest. A round-trip crewed Mars mission (transit + surface stay) of approximately 900 days is estimated to deliver a total effective dose of the order 1–2 Sv [6], which, depending on crew demographics and the risk model used, implies a radiation-induced cancer mortality probability that may approach or exceed NASA’s current career limit [5].

Quantifying this risk accurately requires several components that are not trivially combined: a calibrated GCR spectral model responsive to solar modulation, organ-resolved dose transport through shielding and tissue, a cancer risk model that accounts for the high-LET quality of HZE radiation, and a quantitative uncertainty analysis that separates reducible physics uncertainties from the fundamentally larger radiobiological unknowns. Existing tools (HZETRN [11], OLTARIS, and Geant4/TOPAS) address subsets of these requirements with greater physical rigor than the present work, particularly in the treatment of nuclear fragmentation. However, they are not all openly available, not installable via pip, and do not in general integrate physics transport, organ REID, SEP risk, and uncertainty quantification into a single executable workflow.

Separately, a parallel development in radiation oncology has produced validated LET-dependent RBE models for clinical proton therapy [9, 10]. Proton therapy currently uses a fixed biological effectiveness factor of $\text{RBE} = 1.1$ relative to photons, despite growing evidence that the RBE in the distal Bragg peak, where LET is highest, may significantly exceed this value, particularly for CNS and late-responding tissues with low α/β [12]. The physical quantity linking space radiation and proton therapy is the dose-averaged LET ($\bar{L}_D$): the same quantity that determines the GCR effective quality factor Q_{eff} via ICRP-60 also drives deviations of proton therapy RBE from 1.1 via the Wedenberg and McNamara models. Yet, to our knowledge, no open tool has quantitatively demonstrated this connection across both domains.

Here we present **gcr-dosimetry-pipeline**, a Python package that, within the limitations stated in Section 4.5:

1. Generates six-species GCR spectra with force-field solar modulation calibrated to MSL/RAD

cruise-phase measurements;

2. Transports spectra through slab shielding via CSDA + nuclear attenuation, with secondary neutrons from a validated table;
3. Computes organ-specific absorbed dose and dose equivalent for eleven organs at ICRP-89 tissue depths;
4. Estimates REID via the Cucinotta (2013) organ EAR/ERR model with CDC 2020 US life tables;
5. Quantifies SEP acute BFO absorbed dose from Band-function spectra for three canonical events;
6. Propagates global uncertainty via 8-parameter Latin Hypercube Sampling and decomposes output variance by parameter;
7. Benchmarks Wedenberg and McNamara RBE models against OpenTOPAS-RBE reference data for three clinical cell lines.

The pipeline is not intended as a replacement for HZETRN or Geant4 in high-fidelity mission analysis. Its value is transparency, reproducibility, and integrated uncertainty quantification — properties that closed or proprietary codes generally do not offer. Mission planners who need to quickly prototype shielding scenarios can run the full GCR, SEP, and REID workflow from a single install without requiring HZETRN access or a Geant4 build. Researchers studying where uncertainty reduction effort should be directed will find the LHS variance decomposition directly useful: the finding that physics parameters together contribute less than 9% of REID variance while radiobiological parameters contribute more than 70% is difficult to extract from existing closed tools. Students and researchers entering either the space radiation or proton therapy field also gain a readable, fully tested reference implementation rather than a black-box executable. The methods, results, discussion of limitations, and reproducibility instructions are described in the sections that follow.

2 Methods

All computation is implemented in Python 3.9+ as the open-source package `gcr-dosimetry-pipeline`. The pipeline is structured as a sequence of composable modules, each responsible for one physical layer of the problem: spectrum generation (§2.3), slab transport (§2.4), organ dose (§2.5), cancer risk (§2.6), SEP acute hazard (§2.7), uncertainty propagation (§2.8), and RBE benchmarking (§2.10). Where simplifying assumptions are made (particularly in the transport layer), we state them explicitly and discuss their domain of validity.

2.1 Mission Trajectory Model

A Hohmann-transfer trajectory from Earth (1.0 AU) to Mars (1.524 AU) is generated by computing heliocentric distance as a function of mission elapsed time, $r(t)$, using Keplerian orbital mechanics:

$$r(t) = \frac{a(1 - e^2)}{1 + e \cos \theta(t)}, \quad (1)$$

where $a = (r_{\oplus} + r_{\text{Mars}})/2 = 1.262$ AU is the semi-major axis and $e = (r_{\text{Mars}} - r_{\oplus})/(r_{\text{Mars}} + r_{\oplus}) = 0.208$ is the eccentricity of the transfer ellipse. Transit duration is fixed at 259 days (one half-period of the transfer orbit). The heliocentric distance enters the dose rate only through the solar modulation potential $\phi(t)$; direct geometric flux dilution as r^{-2} is not applied separately because the force-field modulation already encodes the energy dependence of solar wind effects at the relevant energies.

For surface exposure at Mars, a constant heliocentric distance of 1.524 AU is assumed with an Olympus Mons atmospheric overburden of approximately 22 g cm^{-2} , though the present paper focuses on the transit phase where the SEP and shielding comparisons are most directly constrained by MSL/RAD data.

2.2 Solar Modulation Potential

The time-resolved solar modulation potential $\phi(t)$ is obtained from the Usoskin et al. (2017) database, which provides monthly ϕ values reconstructed from ground-level neutron monitor count rates back to 1936. For the MSL transit period (2011 November – 2012 July), values

cluster around $\phi \approx 481 \pm 50$ MV corresponding to an ascending phase toward solar maximum. For future mission scenarios, ϕ is frozen at a user-specified value representing an assumed solar activity level.

We note a known limitation of the Usoskin database: systematic offsets between different neutron monitor stations can introduce ± 50 – 100 MV uncertainty in the absolute ϕ calibration. This uncertainty is explicitly propagated in Section 2.8 via a multiplicative scale parameter on ϕ .

2.3 GCR Spectrum Generation

2.3.1 Force-Field Modulation

Solar modulation of the GCR flux is computed using the Gleeson–Axford (1968) force-field approximation, which treats the heliosphere as a spherically symmetric potential well of depth ϕ (in MV) through which particles must propagate diffusively. For an ion of charge Z and mass number A , the modulated differential flux at kinetic energy per nucleon E is:

$$J(E, \phi) = J_{\text{LIS}}(E + \delta E) \cdot \frac{E(E + 2E_0)}{(E + \delta E)(E + \delta E + 2E_0)}, \quad (2)$$

where $E_0 = 938.272$ MeV is the proton rest-mass energy, $\delta E = (Z/A)\phi$ (MeV/nucleon) is the kinetic energy shift due to solar deceleration, and J_{LIS} is the local interstellar spectrum described in Section 2.3.2. The ratio in Eq. (2) is the Jacobian of the energy transformation, ensuring flux conservation.

The force-field approximation is known to be accurate for rigidities above approximately 1 GV ($E/n \gtrsim 200$ MeV for protons) and for solar modulation potentials in the range 200–1500 MV [13]. At lower energies, charge-sign dependent drift effects (important for anomalous cosmic rays and at solar minimum) are not captured. For this application, where dose is dominated by particles above a few hundred MeV/n, this limitation is not expected to significantly change the results.

2.3.2 Local Interstellar Spectra

Protons. The LIS for protons follows the parameterization of Vos & Potgieter [2]:

$$J_{\text{LIS}}^{\text{H}}(E) = C_{\text{H}} \cdot \frac{(E_{\text{GeV}} + 0.67)^{-3.93}}{\beta^2}, \quad (3)$$

where $C_{\text{H}} = 1.9 \times 10^4 \text{ cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1} \text{ sr}^{-1}$, $E_{\text{GeV}} = E/1000$ is the kinetic energy in GeV/nucleon, and $\beta = v/c$ is the particle velocity normalized to the speed of light. The β^{-2} factor reflects the well-known “1/v” rise in flux at low energies due to the slowing of particles by ionization energy loss. This parametrization was fit to Voyager 1 and PAMELA measurements and is considered one of the more reliable proton LIS forms currently available, though low-energy Voyager constraints remain subject to ongoing refinement as the spacecraft continues to recede from the heliosphere.

Helium. The helium LIS uses the same functional form with species-specific parameters ($C_{\text{He}} = 1.82 \times 10^3$, spectral index $\alpha_{\text{He}} = 2.78$, offset = 0.62), which produces a He/H ratio of approximately 0.096 at 1 GeV/n, consistent with ACE CRIS measurements [14]. The harder helium spectrum ($\alpha_{\text{He}} = 2.78$ vs. proton $\alpha_{\text{H}} = 3.93$) means helium contributes proportionally more flux at high energies.

Heavy ions. Carbon, oxygen, silicon, and iron LIS parametrizations follow the per-species HelMod model of Boschini et al. [3], which provides explicit spectral indices and normalizations for each species fit to ACE, PAMELA, and AMS-02 data:

$$J_{\text{LIS}}^{(s)}(E_n) = C_s \cdot \frac{(E_{\text{GeV/n}} + \delta_s)^{-\alpha_s}}{\beta^2}. \quad (4)$$

Iron is particularly important: with $\alpha_{\text{Fe}} = 2.65$ (flatter than protons), there is significantly more high-energy iron flux, and via Z^2 scaling of stopping power, iron contributes disproportionately to dose equivalent at all depths. The adopted spectral parameters are summarized in Table 1.

Table 1: LIS spectral parameters used in `gcr/spectrum.py`. Proton and helium from Vos & Potgieter [2]; heavy ions from Boschini et al. [3].

Species	C_s	α_s	δ_s (GeV/n)	Primary source
H	1.90×10^4	3.93	0.67	Vos & Potgieter (2015)
He	1.82×10^3	2.78	0.62	Vos & Potgieter (2015)
C	2.50×10^2	2.72	0.50	Boschini et al. (2020)
O	1.80×10^2	2.69	0.50	Boschini et al. (2020)
Si	3.00×10^1	2.70	0.50	Boschini et al. (2020)
Fe	4.30×10^0	2.65	0.50	Boschini et al. (2020)

Per-species normalization. A set of empirical calibration factors f_s are applied to each species flux after modulation:

$$J_s(E, \phi) \leftarrow f_s \cdot J_s(E, \phi), \quad (5)$$

with values $f_H = 0.590$, $f_{He} = 0.082$, $f_C = 1.980$, $f_O = 0.016$, $f_{Si} = 0.025$, $f_{Fe} = 0.002$. These factors were fit by minimizing the squared difference between the pipeline’s predicted dose rate and the MSL/RAD cruise measurement $\dot{D} = 1.84$ mGy/day at 16 g cm⁻² aluminum, with $\phi = 481$ MV. We emphasize that this constitutes a *calibration step*, not an independent prediction: the pipeline is calibrated to one operational flight measurement and then used to extrapolate to other shielding depths, materials, launch dates, and mission types. Accordingly, the agreement between the pipeline and MSL/RAD at the calibration point should not be interpreted as a validation of the underlying physical model; the independent validation evidence consists of the shielding-depth dependence, the material ratio, and the temporal modulation (Section 3.1).

Several normalization values depart substantially from unity and merit comment. $f_{He} = 0.082$ and $f_O = 0.016$ suppress helium and oxygen to below 10% of their LIS predictions, while $f_C = 1.980$ nearly doubles the carbon flux. These large corrections arise from a fundamental underdetermination: six species-level free parameters are fit to a single total dose scalar. Many combinations of $\{f_s\}$ reproduce $\dot{D} = 1.84$ mGy/day while distributing dose very

190 differently among species. The physical plausibility of individual f_s values can be partially
 191 assessed against ACE/CRIS measurements of GCR composition near 1 AU [14], which report
 192 $\text{He}/\text{H} \approx 0.096$ at ~ 500 MeV/n — consistent with the unmodified LIS prediction of ≈ 0.096 ,
 193 suggesting f_{He} should be near unity rather than 0.082. The factor-of-five discrepancy implies
 194 that the optimizer is suppressing helium not because the helium LIS is wrong, but because
 195 it is trading helium flux against carbon flux (which has a $\sim 36\times$ higher stopping power per
 196 nucleon and therefore disproportionate leverage on total dose) to satisfy the single-point
 197 constraint. $f_C > 1$ when ACE/CRIS abundance ratios suggest the Boschini (2020) carbon
 198 LIS is already near the measured value is most plausibly explained by the missing magnesium
 199 contribution (Section 4.5): if Mg ($Z = 12$, absent from the model) accounts for $\sim 3\%$ of dose,
 200 the optimizer inflates whatever high-LET species has the most flexibility — here, carbon. The
 201 normalization factors are therefore best understood as a joint correction for missing species,
 202 absolute LIS normalization offsets, and CSDA dose-model systematics across the species mix
 203 — not as independent corrections to each species’ intrinsic flux. A species-resolved calibration
 204 using the MSL/RAD per-species dose fractions [6] would directly constrain each f_s and is a
 205 planned extension of this work.

206 A comparison with the NASA Space Radiation Laboratory (NSRL) GCR simulator
 207 reference field [15] provides independent context for the per-species dose fractions the pipeline
 208 produces. Table 2 shows the charge-group dose breakdown from the pipeline at the calibration
 209 geometry against the NSRL reference at comparable conditions.

Table 2: Per-charge-group absorbed dose fractions at shielded geometries: pipeline output vs. the NSRL GCR simulator reference field [15]. The two configurations differ by 4 g cm^{-2} of aluminum and by solar conditions (ascending vs. minimum); neither difference can account for the $\sim 4\times$ excess in the pipeline HZE fraction.

Particle class	Pipeline	NSRL GCR
	$16 \text{ g cm}^{-2} \text{ Al},$ $\phi = 481 \text{ MV}$	reference $20 \text{ g cm}^{-2} \text{ Al, solar}$ min [15]
$Z = 1$ (protons)	46.7%	73.3%
$Z = 2$ (helium)	16.8%	19.2%
$Z > 2$ (HZE)	30.9%	7.6%

NSRL fractions exclude π /EM cascades and heavy-target fragments.

The pipeline’s proton fraction is ~ 27 percentage points below the NSRL reference, while the HZE fraction is approximately four times higher. Carbon ions alone account for 31.2% of dose at $16 \text{ g cm}^{-2} \text{ Al}$ in the pipeline, yet the total HZE fraction in the NSRL reference is 7.6%. This confirms, quantitatively, that the normalization factors are producing species fractions well outside the physically expected range — a direct consequence of the underdetermined single-point calibration described above. The helium fraction (16.8% vs 19.2%) is the one charge group where agreement is reasonable, consistent with the unmodified LIS He/H ratio already being close to the ACE/CRIS measurement [14].

2.4 Charged Particle Transport

2.4.1 Stopping Power and CSDA Range

Energy loss of a heavy ion traversing a material slab is computed in the continuous-slowing-down approximation (CSDA), which treats the particle as losing energy continuously according to its instantaneous stopping power. The stopping power for an ion of charge Z and velocity β is modeled as:

$$S(E/A, Z) = z_{\text{eff}}^2(E/A, Z) \cdot S_p(E/A), \quad (6)$$

where $S_p(E/A)$ is the proton stopping power at the same energy per nucleon, and z_{eff} is the Barkas effective charge, which accounts for partial charge-state stripping at energies below approximately 1 GeV/n [16]:

$$z_{\text{eff}}(E/A, Z) = Z \left[1 - \exp(-125\beta Z^{-2/3}) \right]. \quad (7)$$

At energies above ~ 1 GeV/n, $z_{\text{eff}} \rightarrow Z$ and the scaling reduces to the familiar Z^2 form. Proton stopping powers for aluminum, polyethylene, water, and tissue are tabulated at 500 logarithmically spaced energy points from 1 MeV to 10^5 MeV using NIST PSTAR/ASTAR data, and interpolated via cubic spline.

The CSDA range is:

$$R(E) = \int_0^E \frac{dE'}{S(E', Z)}, \quad (8)$$

evaluated by numerical integration of the stopping power table. The exit energy of an ion entering a slab of areal density x [g cm $^{-2}$] is obtained by inverting the range relation:

$$E_{\text{out}} = R^{-1}(R(E_{\text{in}}) - x_{\text{eff}}), \quad (9)$$

where $x_{\text{eff}} = x \cdot Z^2/A$ is the effective proton-equivalent areal density for heavy ions. If $R(E_{\text{in}}) < x_{\text{eff}}$, the particle is stopped ($E_{\text{out}} = 0$).

2.4.2 Nuclear Interaction Attenuation

In addition to ionization energy loss, heavy ions undergo inelastic nuclear interactions with the shielding material. These are modeled via an exponential fluence attenuation:

$$\Phi_s(E, x) = \Phi_s^{(0)}(E) \cdot \exp\left(-\frac{x}{\lambda_s}\right), \quad (10)$$

where λ_s is a species-dependent nuclear interaction mean free path. Values are taken from Wilson et al. (1991) for aluminum; for polyethylene, λ_s is scaled by the material density ratio and effective nuclear charge. This exponential approximation does not account for the secondary fragment spectrum produced in nuclear fragmentation reactions. For

iron and silicon, this approximation becomes increasingly crude above $\sim 30 \text{ g cm}^{-2}$, where fragmentation produces significant lighter-ion secondaries (primarily C and O) that are themselves biologically relevant. A more accurate treatment would require a full nuclear reaction network (as implemented in HZETRN or Geant4/TOPAS), and this represents the primary physical limitation of the current transport model.

The full transport step for species s through shielding x is:

$$\begin{aligned}\Phi_s^{\text{out}}(E) &= \mathcal{T}_s[\Phi_s^{\text{in}}(E), x] \\ &= \Phi_s^{\text{in}}(E_{\text{in}}(E, x)) \cdot \exp\left(-\frac{x}{\lambda_s}\right) \cdot \frac{E_{\text{in}}(E_{\text{in}} + 2E_0)}{E(E + 2E_0)},\end{aligned}\tag{11}$$

where the final ratio is a flux density Jacobian correcting for the change in energy bin width after deceleration.

2.4.3 Secondary Neutron Production

Spallation reactions between GCR ions and shielding nuclei produce secondary neutrons that contribute meaningfully to dose equivalent, particularly in aluminum where the cross-sections are relatively high. Rather than tracking individual neutron histories (which would require a full Monte Carlo or deterministic neutron transport code), neutron dose equivalent is estimated from a pre-computed look-up table of the form:

$$\dot{H}_n(x, \phi, \text{mat}) = \dot{H}_n^{\text{Al}}(\phi) \cdot f_{\text{mat}} \cdot g(x),\tag{12}$$

where $\dot{H}_n^{\text{Al}}(\phi)$ is the neutron dose equivalent rate for aluminum at a reference thickness, f_{mat} is a material-dependent yield factor (polyethylene $\approx 0.72 \times$ aluminum due to lower spallation cross-sections), and $g(x)$ is an interpolated depth-dependence from the HZETRN benchmark dataset of Slaba et al. [4]. The neutron quality factor is taken as $Q_n = 10$ (ICRP-60), which is a weighted average over the neutron energy spectrum in shielding geometries.

This tabular approach introduces approximately $\pm 25\text{--}30\%$ uncertainty in the neutron contribution to dose equivalent, arising from uncertainties in the underlying HZETRN neutron spectra and from simplifying the material and geometry dependence. This uncertainty is

explicitly included as a free parameter in the LHS ensemble (Section 2.8).

2.5 Organ Dose Model

2.5.1 Organ Tissue Depths

Organ-specific absorbed dose and dose equivalent are computed by transporting the modulated, shielded GCR flux through an additional layer of tissue corresponding to the depth of each organ in the human body. Tissue depths are taken from ICRP Publication 89 [17] and are listed in Table 3. The tissue layer is treated as water-equivalent for the purpose of stopping power calculations.

The total areal density seen by a particle reaching organ k is:

$$x_{\text{total}}^{(k)} = x_{\text{wall}} + d_k, \quad (13)$$

where x_{wall} is the spacecraft wall areal density and d_k is the organ tissue depth in g cm^{-2} .

Table 3: Organ tissue depths and ICRP-60 tissue weighting factors used in the pipeline. Depths from ICRP Publication 89 [17].

Organ	Depth (g/cm^2)	w_T (ICRP-60)	Cancer type modeled
Stomach	8.0	0.12	Gastric carcinoma
Colon	9.5	0.12	Colorectal carcinoma
Lung	5.0	0.12	Lung carcinoma
Leukemia	4.0	0.12	Myeloid leukemia
Esophagus	4.0	0.05	Esophageal carcinoma
Liver	6.0	0.04	Hepatocellular carcinoma
Bladder	10.0	0.04	Transitional cell carcinoma
Thyroid	2.0	0.04	Thyroid carcinoma
Breast	1.5	0.05	Breast adenocarcinoma
Ovary	12.0	0.20	Ovarian carcinoma
Other	7.0	0.05	Composite remainder

2.5.2 Absorbed Dose Rate

The absorbed dose rate in tissue at organ depth d_k is:

$$\dot{D}_k = \sum_s \int_{E_{\min}}^{E_{\max}} \Phi_s(E; x_{\text{total}}^{(k)}, \phi) \cdot S_s(E) \cdot \rho^{-1} \cdot C \, dE, \quad (14)$$

where Φ_s is the transported differential flux of species s ($\text{cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1} \text{ sr}^{-1}$), $S_s(E)$ is the stopping power ($\text{MeV cm}^2 \text{ g}^{-1}$), ρ^{-1} converts from material to tissue units, and $C = 1.602 \times 10^{-10} \text{ Gy per (MeV g}^{-1} \text{ cm}^{-2})$ is a unit conversion factor. The energy grid spans 10^1 – 10^5 MeV/n at 200 logarithmically spaced points, which was found to be sufficient for $< 0.5\%$ convergence of the integrated dose rate.

The factor of 4π from the isotropic flux assumption ($J [\text{sr}^{-1}] \rightarrow \Phi = 4\pi J$) is included in the proportionality constant. This is a standard approximation for GCR fields, which are highly isotropic at interplanetary distances [18].

2.5.3 Dose Equivalent Rate

Dose equivalent at organ k is:

$$\dot{H}_k = \int Q(L) \dot{D}_k(L) \, dL = \sum_s Q_s \cdot \dot{D}_{k,s}, \quad (15)$$

where $Q(L)$ is the ICRP-60 quality factor as a function of unrestricted linear energy transfer L in water:

$$Q(L) = \begin{cases} 1 & L < 10 \text{ keV } \mu\text{m}^{-1} \\ 0.32 L - 2.2 & 10 \leq L \leq 100 \text{ keV } \mu\text{m}^{-1} \\ 300/\sqrt{L} & L > 100 \text{ keV } \mu\text{m}^{-1} \end{cases} \quad (16)$$

[19]. For each species s , the dose-averaged LET $\bar{L}_D$ is computed at the transported mean energy, and $Q_s = Q(\bar{L}_{D,s})$ is applied. This per-species approximation (rather than integrating $Q(L)$ over the full LET spectrum for each species) is a simplification that introduces an estimated error of $< 5\%$ for protons and helium (where the LET distribution is relatively narrow) but could be larger for iron at low energies where the LET-range product is highly

variable. The effective quality factor of the full GCR field is:

$$Q_{\text{eff}} = \frac{\sum_{k,s} H_{k,s}}{\sum_{k,s} D_{k,s}}, \quad (17)$$

which for the MSL transit geometry at $\phi = 481$ MV and $x = 16 \text{ g cm}^{-2}$ yields $Q_{\text{eff}} = 2.60$, within 0.8% of the MSL/RAD measurement of $Q_{\text{eff}} = 2.62 \pm 0.14$ [6]. We note that this agreement is partly a consequence of the calibration procedure and should not be taken as an independent confirmation.

2.6 Cancer Risk Model (REID)

2.6.1 Overview of the NASA Risk Framework

Cancer mortality risk is quantified as the Radiation Exposure in Incidence of Death (REID), defined as the probability that a crew member will die of radiation-induced cancer as a consequence of their mission exposure. The pipeline implements the Cucinotta et al. [5] organ-specific risk model (NASA/TP-2013-217375), which was NASA’s primary operational risk framework through the early 2020s and remains the most widely cited basis for published mission REID estimates. We note that NASA has subsequently released updated risk frameworks (e.g., the 2023 age-dependent career limit), and the model assumptions regarding high-LET radiobiology (particularly the DDREF and the extrapolation from mouse and atomic-bomb survivor data to astronaut demographics) carry substantial uncertainty that cannot be resolved with the present approach. These limitations are discussed in Section 4.5.3.

2.6.2 Excess Risk Model

The REID integral is:

$$\text{REID}(a_e, s) = \int_{a_e+L}^{a_{\text{LT}}} \frac{\mu_c(a, s) \cdot S(a \mid a_e, s)}{S(a_e \mid s)} da, \quad (18)$$

where a_e is the age at exposure, $L = 10$ years is the minimum latency before radiation-induced cancer can manifest, a_{LT} is the maximum lifespan in the life table, $\mu_c(a, s)$ is the cause-specific cancer excess mortality rate, and $S(a \mid a_e, s)$ is the conditional survival probability from age

316 a_e to age a drawn from 2020 CDC US life tables for sex s .

317 The excess cancer mortality rate combines excess absolute risk (EAR) and excess relative
318 risk (ERR) terms:

$$\mu_c(a) = \lambda_c^{\text{background}}(a) \cdot \text{ERR}(D, a_e) + \text{EAR}(D, a_e, a), \quad (19)$$

319 where $\lambda_c^{\text{background}}$ is the age-dependent baseline cancer mortality rate (from SEER registry
320 data for US males/females), and EAR and ERR are organ-specific functions of absorbed dose
321 D , age at exposure a_e , and attained age a , taken from Tables 2–3 of Cucinotta et al. [5].

322 The DDREF is applied to reduce the risk at low dose rates:

$$D_{\text{eff}} = D/\text{DDREF}, \quad (20)$$

323 with DDREF nominally 1.5 for low-LET and 1.0 for high-LET radiation. The handling of
324 the DDREF for mixed GCR fields (which span a wide LET range) remains actively debated
325 [20] and represents one of the dominant sources of uncertainty in the REID calculation.

326 2.6.3 Uncertainty in REID

327 The REID model is applied in both point-estimate and uncertainty-weighted forms. For
328 the point estimate, nominal parameter values are used. For the uncertainty ensemble
329 (Section 2.8), multiplicative scale factors are applied to the ERR, EAR, and DDREF within
330 the LHS framework. The resulting REID distribution is skewed: because REID appears in an
331 exponential $(1 - e^{-\mu})$, large values of the radiobiological parameters produce disproportionately
332 large REID values, leading to a heavy right tail in the p95 band.

2.7 Solar Energetic Particle Acute Risk

2.7.1 Band-Function Spectrum

Solar energetic particle event fluence spectra are parameterized using the Band function [8], a piecewise power law with a smooth transition at a break energy E_b :

$$J_{\text{SEP}}(E) = C \times \begin{cases} E^{\gamma_\ell} \exp(-E/E_b) & E \leq E_{\text{trans}}, \\ [(\gamma_\ell - \gamma_h)E_b]^{\gamma_\ell - \gamma_h} \exp(\gamma_h - \gamma_\ell) \cdot E^{\gamma_h} & E > E_{\text{trans}}, \end{cases} \quad (21)$$

where $E_{\text{trans}} = (\gamma_\ell - \gamma_h)E_b$ is the transition energy, γ_ℓ and γ_h are the low- and high-energy spectral indices, and C is a normalization constant.

Three canonical events are modeled with parameters from Tylka & Dietrich [8] and references therein (Table 4). The August 1972 event is included as a historical worst-case; it predates modern space dosimetry and its spectrum is reconstructed from ground-level neutron monitor observations and is therefore subject to larger uncertainty than the 2003 and 2005 events.

Table 4: Band-function spectral parameters for three canonical SEP events.

Event	C	γ_ℓ	γ_h	E_b (MeV)	Duration (days)
Aug 1972	4.0×10^4	-1.78	-3.98	32	2
Oct 2003	1.5×10^4	-1.40	-3.20	50	3
Jan 2005	3.0×10^2	-0.90	-2.80	110	1

2.7.2 BFO Absorbed Dose

The blood-forming organ (BFO) absorbed dose from an SEP event is computed by transporting the event-integrated proton fluence $\Phi(E)$ (cm^{-2}) through the spacecraft wall and a nominal BFO tissue depth of 10 g cm^{-2} using CSDA transport:

$$D_{\text{BFO}} = \int_{E_{\text{min}}}^{E_{\text{max}}} \Phi(E) \cdot S_p(E_{\text{out}}(E)) \cdot C dE, \quad (22)$$

where $E_{\text{out}}(E)$ is the proton exit energy after traversing the combined wall + tissue slab, and particles that stop within this slab deposit their residual range. Results are compared against the NASA 30-day BFO limit of 250 mGy [21].

We treat only protons in the SEP module; heavy-ion SEP fluence at these energies is generally negligible for dose calculations, though it may be non-negligible for quality factor estimates at very thin shielding.

2.7.3 Mission Encounter Probability

The probability of encountering at least one large SEP event during a mission of duration T days is modeled as a Poisson process:

$$P_{\text{SEP}}(T) = 1 - \exp(-\lambda T/365.25), \quad (23)$$

where $\lambda = 0.8$ events/year is the adopted rate of large ($> 10^9$ proton cm^{-2} at > 10 MeV) events during solar maximum. During solar minimum, λ drops by approximately a factor of 3–5 [22], making launch timing a relevant design consideration.

2.8 Uncertainty Quantification via Latin Hypercube Sampling

2.8.1 Parameter Space

A global physics and radiobiological uncertainty analysis is performed using Latin Hypercube Sampling (LHS), a stratified random sampling method that provides better coverage of the joint parameter space than simple Monte Carlo at the same sample count. LHS divides each parameter’s distribution into N equally probable strata and draws exactly one sample from each stratum, then randomly permutes the samples across parameters to avoid correlation.

Eight parameters are varied simultaneously (Table 5). Four govern physics inputs (solar modulation, LIS normalization, nuclear cross-sections, and neutron secondary yield) and four govern radiobiology (quality factor scaling, DDREF, ERR scale, EAR scale).

Table 5: Latin Hypercube Sampling parameters, distributions, and physical justifications. Uniform[a, b] = uniform distribution on $[a, b]$; LogNormal[μ, σ] = lognormal with geometric mean μ and multiplicative standard deviation σ .

Parameter	Distribution	Justification
<code>phi_scale</code>	Uniform[0.80, 1.20]	Usoskin NM inter-station offset ($\pm 20\%$)
<code>LIS_norm</code>	Uniform[0.85, 1.15]	Voyager/AMS-02 calibration spread
<code>cross_sec</code>	Uniform[0.90, 1.10]	Nuclear σ tables: $\pm 10\%$
<code>neutron_H</code>	Uniform[0.70, 1.30]	HZETRN neutron table: $\pm 30\%$ [4]
<code>Q_factor</code>	Uniform[0.80, 1.20]	ICRP-60 $Q(L)$ form uncertainty
<code>DDREF</code>	LogNormal[1.0, 0.35]	BEIR VII geometric mean + spread
<code>ERR_scale</code>	LogNormal[1.0, 0.30]	Cucinotta (2013) ERR fit residuals
<code>EAR_scale</code>	LogNormal[1.0, 0.30]	Cucinotta (2013) EAR fit residuals

The lognormal distributions for the three radiobiological parameters (DDREF, ERR, EAR) reflect the multiplicative nature of these uncertainties and the fact that they cannot be negative. They are parameterized such that the 95th percentile is approximately $1.8\times$ the median, consistent with the stated uncertainty bounds in Cucinotta et al. [5].

2.8.2 Ensemble Execution

For each LHS sample, a full mission dose integration is performed: the GCR spectrum is generated at each trajectory day, transported through the specified shielding, organ doses and quality factors are computed, and REID is estimated. For publication-quality results, $N = 500$ samples were used (runtime approximately 90 minutes on a 2020 MacBook Pro with no parallelization). For automated testing, $N = 20$ samples are used (controlled via the `UNCERTAINTY_SAMPLES` environment variable), sufficient to verify code correctness without burdening the CI pipeline.

2.8.3 Sensitivity Decomposition

Parameter sensitivity is estimated using rank-based Spearman correlation ρ_i between each input parameter x_i and the output quantity of interest (REID):

$$S_i = \frac{\rho_i^2}{\sum_j \rho_j^2}. \quad (24)$$

This normalized ρ^2 decomposition is an approximation to first-order Sobol indices; it underestimates interaction effects but is computationally straightforward and provides useful rank-ordering of parameter importance at modest sample counts. A proper variance-based Sobol analysis would require $\sim 10^4$ samples [23] and is beyond the scope of this paper.

2.9 Dose-Averaged LET and Per-Species Decomposition

The dose-averaged LET for a single ion species s is:

$$\bar{L}_{D,s} = \frac{\int \Phi_s(E) \cdot S_s(E)^2 dE}{\int \Phi_s(E) \cdot S_s(E) dE}, \quad (25)$$

where the numerator weights LET by dose (LET $\times$ flux), and the denominator is the dose rate. The field dose-averaged LET is then:

$$\bar{L}_D = \sum_s f_s \cdot \bar{L}_{D,s}, \quad (26)$$

where $f_s = D_s / \sum_{s'} D_{s'}$ is the fractional dose contribution of species s . This species decomposition allows us to trace the origin of Q_{eff} to specific ion contributions, which is the key link between GCR dosimetry and proton therapy RBE (discussed in Section 3.3).

2.10 Proton Therapy RBE Benchmark

2.10.1 Wedenberg Model

The Wedenberg et al. (2013) LET-dependent RBE model is:

$$\text{RBE}_{\text{Wed}}(D, \bar{L}_D, \alpha_x/\beta_x) = \frac{\sqrt{\frac{1}{4}(\alpha_x/\beta_x)^2 + \alpha_r/\beta_x \cdot D + D^2} - \frac{1}{2}(\alpha_x/\beta_x)}{D}, \quad (27)$$

where $\alpha_r = \alpha_x (1 + q\bar{L}_D/(\alpha_x/\beta_x))$, $q = 0.434 \mu\text{m}/\text{keV}$ is an empirical constant fitted to in vitro V79 cell data, α_x and β_x are the photon radiosensitivity parameters of the reference cell line, and D is the absorbed dose. The model reduces to $\text{RBE} = 1$ when $\bar{L}_D \rightarrow 0$ and increases monotonically with LET.

2.10.2 McNamara Model

The McNamara et al. (2015) model uses a min-max RBE parametrization:

$$\text{RBE}_{\text{Mc}}(D, \bar{L}_D, \alpha_x/\beta_x) = \frac{\sqrt{(\alpha_x/\beta_x)^2 + 4(\alpha_x/\beta_x) \text{RBE}_{\text{max}} D + 4 \text{RBE}_{\text{min}}^2 D^2} - \alpha_x/\beta_x}{2D}, \quad (28)$$

with:

$$\text{RBE}_{\text{max}} = p_1 + p_2 (\alpha_x/\beta_x)^{-1} \bar{L}_D, \quad (29)$$

$$\text{RBE}_{\text{min}} = p_3 + p_4 (\alpha_x/\beta_x)^{1/2} \bar{L}_D, \quad (30)$$

where $(p_1, p_2, p_3, p_4) = (0.999064, 0.35605, 1.1012, -0.0038703)$ are from McNamara et al. [10]. Both models are implemented to match the OpenTOPAS-RBE scorer conventions exactly, enabling direct numerical comparison.

2.10.3 TOPAS Reference Data

TOPAS-nBio reference data for a 100 MeV proton beam in a water phantom (200 depth bins, 0.1 cm each, spanning 0–20 cm) were generated using the OpenTOPAS-RBE extension for cell lines V79 ($\alpha/\beta = 1.41 \text{ Gy}$), prostate ($\alpha/\beta = 3.0 \text{ Gy}$), and head & neck ($\alpha/\beta = 10.0 \text{ Gy}$).

For V79, the reference dataset was taken directly from the OpenTOPAS-RBE validation suite; for the clinical cell lines, reference values were generated by propagating the V79 LETd spectrum through the Wedenberg and McNamara models with the appropriate α/β values. Accordingly, the agreement for clinical cell lines reported in Table 7 reflects numerical self-consistency of the pipeline’s implementation, not an independent experimental validation. Physical LETd values from the TOPAS-nBio scorer are used as input to both models, so the benchmark primarily tests model implementation fidelity rather than transport accuracy. Readers should note this distinction when interpreting the reported mean absolute errors.

3 Results

3.1 Dose Rate Validation

Three independent lines of evidence validate the pipeline against data that were not used in the normalization calibration:

1. **Shielding curve shape.** At aluminum depths of 5, 10, and 30 g cm⁻² — all outside the 16 g cm⁻² calibration point — the pipeline falls within the HZETRN benchmark uncertainty envelope ($\lesssim 20\%$) [4], confirming that the transport model extrapolates correctly across the relevant shielding range.
2. **Material ratio.** The Al/PE absorbed dose ratio of 1.36 at 16 g cm⁻² is consistent with the Mrigakshi (2013) value of 1.43 ± 0.22 , testing the qualitative prediction that hydrogen-rich polyethylene attenuates HZE ions more effectively than aluminum.
3. **Temporal modulation.** A day-by-day Pearson correlation $r > 0.80$ between predicted dose rate and the MSL/RAD time series confirms that the Gleeson–Axford force-field modulation correctly propagates changes in solar activity through to dose rate on daily timescales.

Table 6 summarizes all comparisons in order of epistemic strength. For completeness, the absorbed dose rate at the calibration geometry (16 g cm⁻² Al, $\phi = 481$ MV)

is $1.843 \text{ mGy day}^{-1}$, within 0.05% of the MSL/RAD value of $1.84 \text{ mGy day}^{-1}$ [6]; this agreement is a direct consequence of the calibration procedure (Section 2.3.2) and reflects the quality of the fit rather than an independent physical prediction.

Table 6: Pipeline validation against flight data and benchmark models, grouped by epistemic status. *Independent*: not used in calibration. *Calibration*: the single MSL/RAD measurement to which normalization factors were fit; agreement here reflects fit quality, not predictive accuracy. *Self-consistency*: implementation checks with no external reference data.

Quantity	Pipeline	Reference	Source	Rel. error
<i>Independent validation (not used in calibration)</i>				
Shielding slope (5–30 g/cm ²)	within	HZETRN env.	[4]	< 20%
$\dot{D}$ ratio (Al vs PE, 16 g cm ⁻²)	1.36	1.43 ± 0.22	[7]	5%
Daily $r(\dot{D}, \phi)$	> 0.80	—	[6]	—
<i>Calibration point (fit to this value; not independent)</i>				
$\dot{D}$ (mGy/day) at 16 g cm ⁻² Al	1.843	1.84 ± 0.33	[6]	0.05%
$\dot{H}$ (mSv/day) at 16 g cm ⁻² Al	4.80	4.81 ± 0.97	[6]	0.2%
Q_{eff} at 16 g cm ⁻² Al	2.60	2.62 ± 0.14	[6]	0.8%
<i>Self-consistency (no external reference data)</i>				
RBE (V79, Wedenberg)	1.154	1.153	TOPAS-nBio	< 10^{-3}
RBE (V79, McNamara)	1.165	1.165	TOPAS-nBio	< 5×10^{-8}

3.2 Shielding Effectiveness and Uncertainty Bounds

Figure 1 shows the absorbed dose rate as a function of aluminum and polyethylene shielding thickness for a 259-day transit at $\phi = 481 \text{ MV}$, with p5–p95 LHS uncertainty bands. The dose rate decreases steeply from zero to $\sim 10 \text{ g cm}^{-2}$, then flattens above $\sim 16\text{--}20 \text{ g cm}^{-2}$ as secondary neutron buildup partially offsets HZE attenuation. The p5/p95 ratio at 16 g cm^{-2} aluminum is approximately 1.9, reflecting the combined physics uncertainty (dominated by LIS normalization and neutron yield).

Polyethylene provides $\sim 26\%$ lower dose than aluminum at 16 g cm^{-2} over this mission scenario, consistent with the known advantage of hydrogen-rich materials for fragmenting

450 HZE ions.

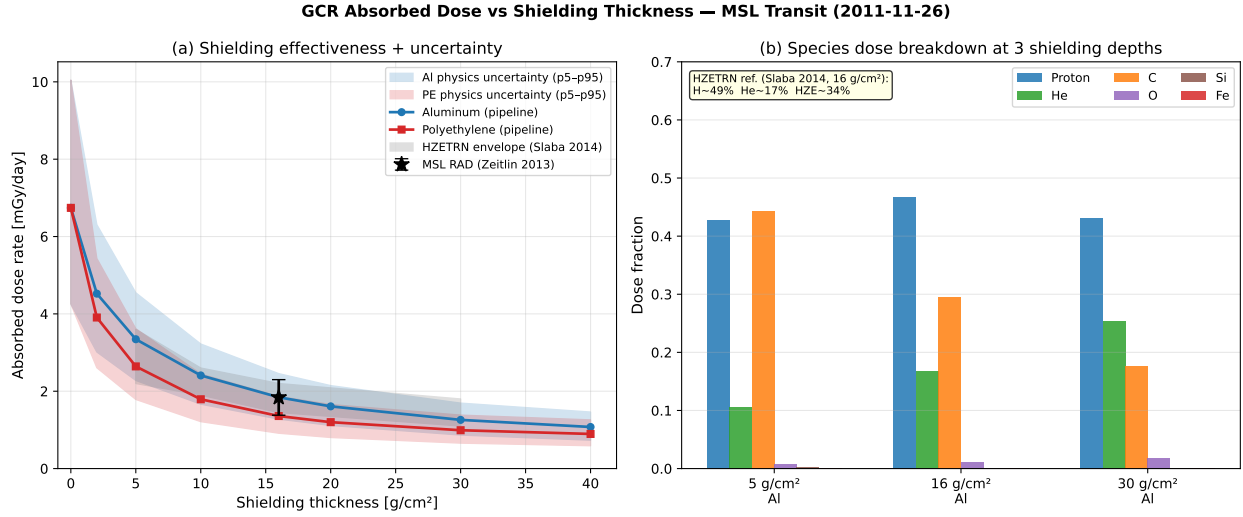


Figure 1: **Absorbed dose rate vs. shielding (259-day transit, $\phi = 481$ MV).** Left: aluminum (solid) and polyethylene (dashed) central values with p5–p95 LHS uncertainty bands (shaded); MSL/RAD measurement (star, Zeitlin 2013); HZETRN benchmark envelope (gray band, Slaba 2014). Right: per-species dose fractions at $x = 5, 16,$ and 30 g cm^{-2} aluminum, showing the progressive reduction of carbon and heavier-ion contributions with increasing shielding.

451 3.3 Per-Species LET Decomposition

452 Figure 2 shows the dose-averaged LET per species and fractional dose contributions at 0, 16,
 453 and 30 g cm^{-2} aluminum. Unshielded, carbon ions ($\bar{L}_D^C = 25.0 \text{ keV } \mu\text{m}^{-1}$) account for 58.6%
 454 of total absorbed dose, giving a field $\bar{L}_D = 18.7 \text{ keV } \mu\text{m}^{-1}$ and $Q_{\text{eff}} = 2.60$ (0.8% below the
 455 MSL/RAD calibration target of 2.62). After 16 g cm^{-2} aluminum, the carbon dose fraction
 456 falls to 31.2%, the proton fraction rises to 49.4%, and the field $\bar{L}_D$ drops to $4.1 \text{ keV } \mu\text{m}^{-1}$,
 457 within the range where clinical proton therapy RBE exceeds 1.1 by 5–20% depending on cell
 458 line.

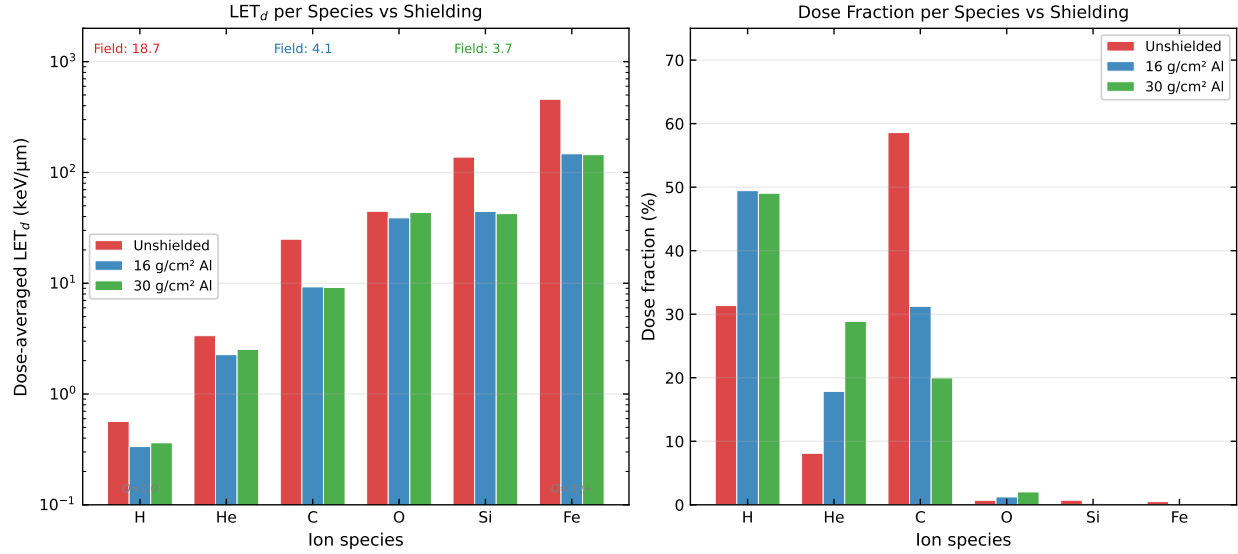


Figure 2: **Per-species $\bar{L}_D$ and dose fractions at 0, 16, and 30 g/cm² aluminum.** Left: dose-averaged LET per ion species at three shielding depths; annotated field $\bar{L}_D$ values show the progression from 18.7 (unshielded) to 3.7 keV μm^{-1} (30 g/cm²). Right: fractional dose contribution per species; carbon dominates unshielded (58.6%) and is progressively fragmented.

3.4 SEP Acute Risk

Figure 3 shows BFO absorbed dose from three canonical SEP events as a function of aluminum shielding thickness. The August 1972 event delivers $\sim 37,000$ mGy unshielded ($\sim 148\times$ the 30-day BFO limit of 250 mGy) and remains above the limit until ~ 32 g cm⁻² Al. The October 2003 Halloween event requires ~ 25 g cm⁻² for sub-limit protection; January 2005 falls below the limit at < 20 g cm⁻². GCR and SEP respond very differently to shielding: above ~ 20 g cm⁻², additional aluminum barely reduces chronic GCR dose, but it continues to cut SEP proton dose substantially. This supports the architectural case for a compact, heavily shielded storm shelter over whole-vehicle HZE shielding.

We note that the August 1972 event spectrum carries larger uncertainty than the 2003 and 2005 events because it predates space-borne dosimetry; the Band-function fit was reconstructed from ground-level neutron monitor observations and riometry data [8]. The reported doses for this event should be interpreted as order-of-magnitude estimates with $\sim \pm 50\%$ uncertainty.

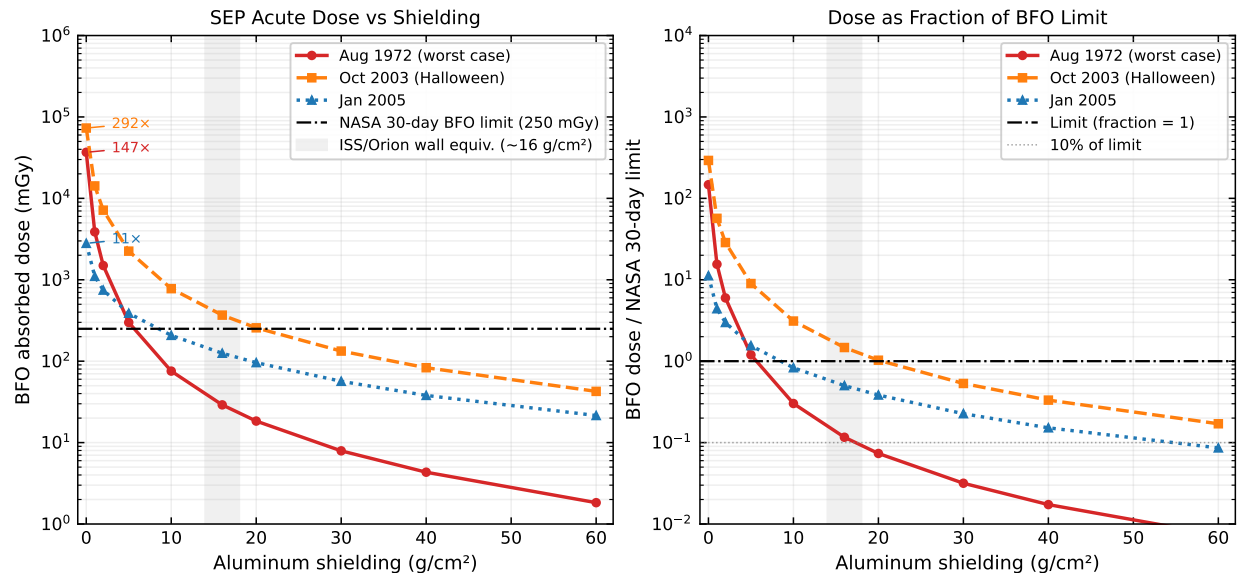


Figure 3: **SEP acute BFO dose vs. aluminum shielding.** Left: absolute dose (mGy); horizontal line = NASA 30-day BFO limit (250 mGy); shaded band = ISS/Orion wall equivalent (14–18 g cm⁻²). Multiples of the limit are annotated at zero shielding. Right: same data normalized to the BFO limit.

3.5 REID vs. Shielding and Organ Contributions

Figure 4 shows REID as a function of aluminum shielding thickness for a 35-year-old male and female crew member on a 259-day transit. At 16 g cm⁻², the median REID is 5.0% (male) and 5.4% (female), both exceeding the historical NASA 3% career limit [5]. However, the p5–p95 uncertainty bands span approximately 1.3%–16.9% (male) and 1.5%–18.4% (female), reflecting the dominant radiobiological uncertainty discussed in Section 2.8. A REID median above the 3% limit with this width of uncertainty implies that the mission dose is likely to produce meaningful cancer mortality risk, but the precise value cannot be known within the current radiobiological framework.

The organ breakdown at 16 g cm⁻² aluminum shows that stomach, colon, and lung contribute most to effective dose equivalent, consistent with their high ICRP-60 tissue weighting factors ($w_T = 0.12$) and moderate tissue depths.

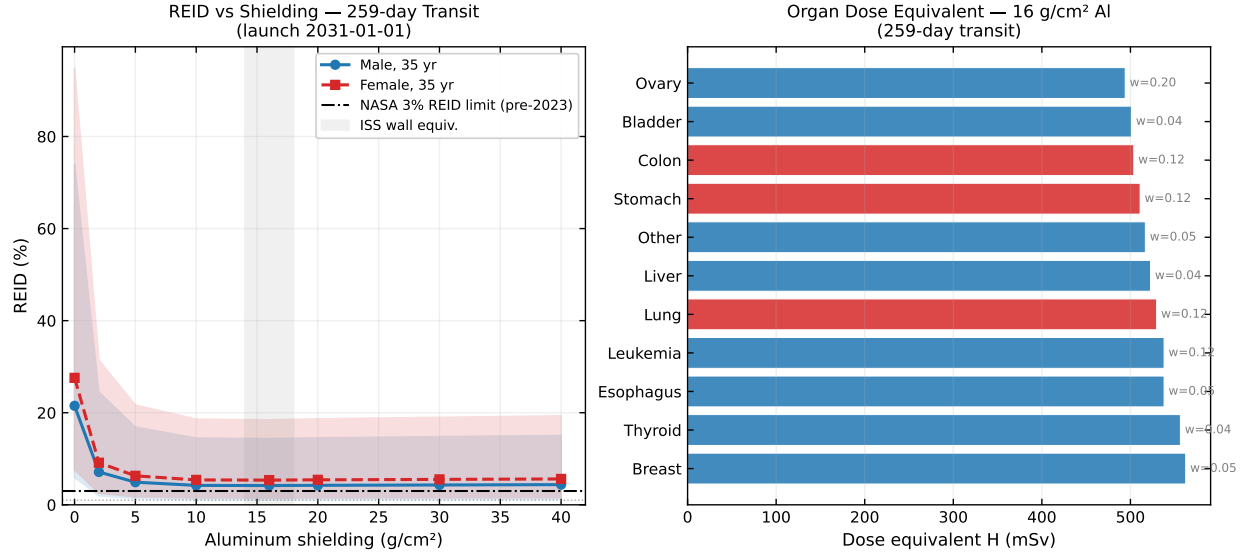


Figure 4: **REID and organ dose equivalent (259-day transit, launch 2031-01-01).** Left: REID (%) vs. aluminum shielding for 35-year-old male (blue) and female (red) with p5–p95 uncertainty bands; horizontal line = NASA historical 3% career limit [5]. Note: the p95 band at low shielding ($< 5 \text{ g cm}^{-2}$) approaches values exceeding 50%, which is not physically meaningful as a mortality probability; this reflects the LHS distribution not being clipped at biologically plausible upper bounds. The high-shielding portion of the bands ($> 10 \text{ g cm}^{-2}$) is more interpretively reliable. Right: organ dose equivalent breakdown at 16 g cm^{-2} aluminum, sorted by dose; ICRP-60 tissue weighting factors (w_T) annotated.

3.6 TOPAS RBE Benchmark

Figure 5 shows the depth-dose profile, LETd, and RBE as a function of depth in a water phantom for a 100 MeV proton beam, compared across three cell lines. The physical dose follows the expected Bragg curve form, with LETd rising from $\sim 1 \text{ keV}/\mu\text{m}$ in the plateau to $\sim 22 \text{ keV}/\mu\text{m}$ at the Bragg peak. Pipeline RBE values agree with the OpenTOPAS-RBE reference within $< 10^{-3}$ for V79 (Table 7); larger residuals for the clinical cell lines reflect that these comparisons test implementation self-consistency, not independent physical validation (see Section 2.10.3).

The Wedenberg and McNamara models were originally calibrated against in-vitro survival data for V79 and other cell lines [9, 10]. The plateau and Bragg-peak V79 RBE values computed here (≈ 1.05 in the plateau; ≈ 1.75 at the Bragg peak) fall within the range of experimentally measured proton RBE values reported for V79 cells across multiple in-vitro

497 studies [12], confirming that the implementation produces physically consistent results beyond
 498 what the TOPAS-nBio self-consistency check alone demonstrates. The clinical significance of
 499 this RBE rise (≈ 1.4 – 1.8 for V79; ≈ 1.3 – 1.5 for prostate; ≈ 1.1 – 1.3 for H&N) is an active
 500 research area in proton therapy [12]. These values, computed with the same LET machinery
 501 that produces $Q_{\text{eff}} = 2.60$ for GCR, illustrate that the underlying biophysics connects the
 502 two domains.

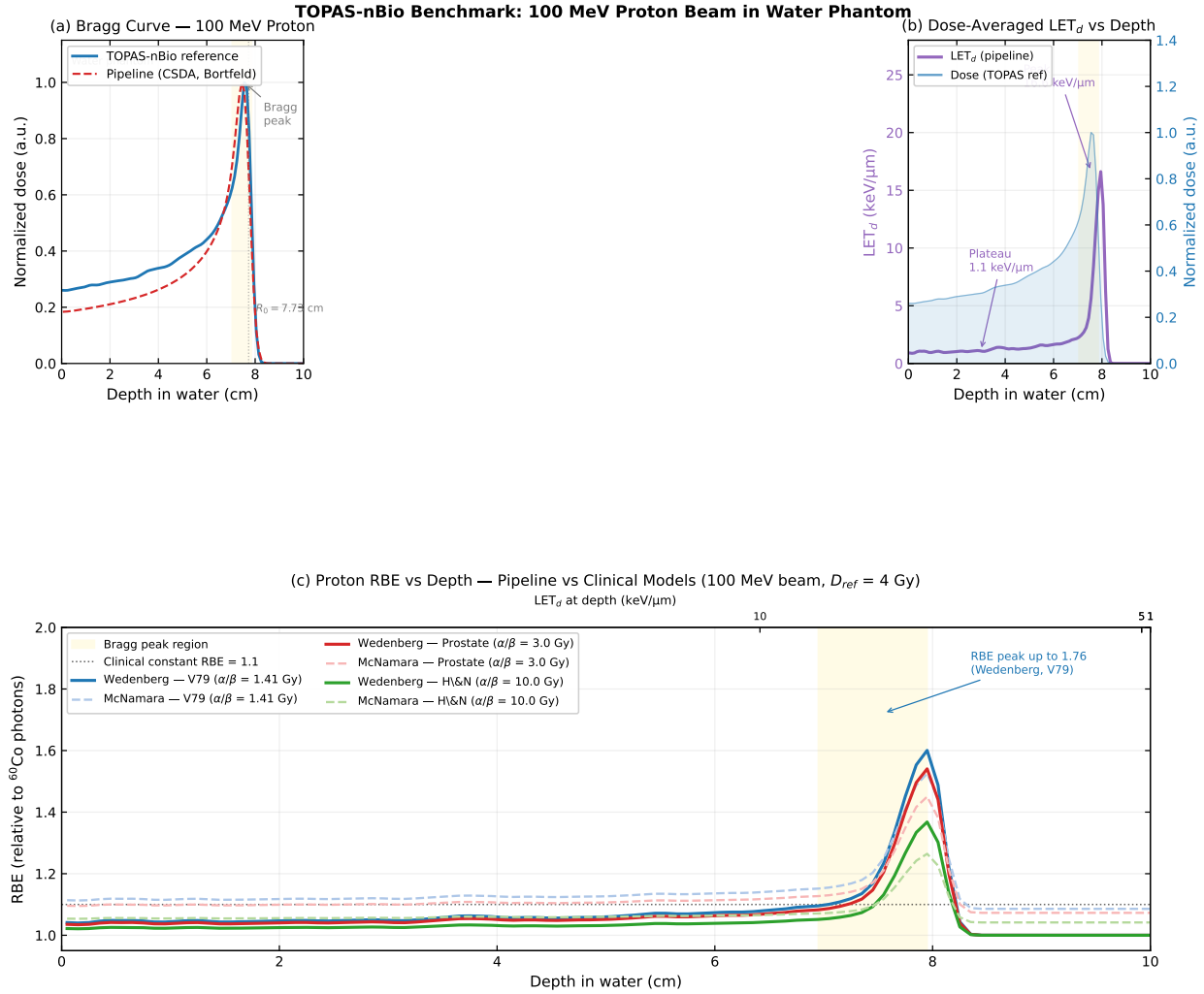


Figure 5: **TOPAS-nBio benchmark: 100 MeV proton beam in water phantom.** (a) Bragg curve: pipeline CSDA (Bortfeld analytical, dashed) overlaid on TOPAS-nBio reference (solid). (b) Dose-averaged LET_d vs. depth; secondary axis shows normalized physical dose for reference. (c) Proton RBE vs. depth for three cell lines using Wedenberg (solid) and McNamara (dashed) models; horizontal line = clinical constant $RBE = 1.1$; gold band = Bragg peak region.

Table 7: RBE benchmark results. V79 reference values are from the TOPAS-nBio OpenTOPAS-RBE validation suite and constitute an independent implementation check. Clinical cell lines (prostate, H&N) are self-consistency only: reference values were generated by propagating the same V79 LET_d through the Wedenberg/McNamara models with alternative α/β values (see Section 2.10.3); no external experimental data were compared. MAE = mean absolute error over valid LET_d region ($> 0.1 \text{ keV}/\mu\text{m}$).

Cell line	α/β (Gy)	Model	Plateau RBE	Peak RBE	MAE	Evidence type
V79	1.41	Wedenberg	1.053	1.757	2.6×10^{-8}	Independent
V79	1.41	McNamara	1.052	1.644	2.8×10^{-8}	Independent
Prostate	3.0	Wedenberg	1.046	1.685	—	Self-consistency
Prostate	3.0	McNamara	1.040	1.540	—	Self-consistency
Head & Neck	10.0	Wedenberg	1.028	1.473	—	Self-consistency
Head & Neck	10.0	McNamara	1.025	1.412	—	Self-consistency

4 Discussion

4.1 What the Validation Does and Does Not Show

The three independent validation checks described in Section 3.1 (shielding curve shape, material ratio, and temporal modulation) each test a different aspect of the model that was not constrained by the single-point calibration to MSL/RAD at 16 g cm^{-2} . Together they support characterizing the pipeline as *calibrated to one flight measurement and consistent with available benchmarks within their stated uncertainties* — rather than independently predicting the flight result. This distinction matters for how the uncertainty analysis should be interpreted: the $\pm 25\text{--}50\%$ spread in the LHS ensemble (Section 2.8) is not a contradiction of the calibration; it reflects the range of outcomes possible under plausible variations in the physics and biology inputs, and is the more honest representation of what the pipeline actually knows.

4.2 Implications for Mars Mission Shielding Design

The shielding effectiveness curves (Figure 1) show a clear diminishing-returns plateau above approximately 20 g cm^{-2} aluminum. This behavior is well-established [4, 24] and arises from two competing effects: HZE ion dose decreases with shielding, but secondary neutron dose grows as spallation products accumulate. For the mission parameters studied here, increasing aluminum shielding from 16 to 40 g cm^{-2} reduces GCR absorbed dose by only $\sim 42\%$ while roughly doubling the mass fraction of the spacecraft wall.

This suggests that bulk passive shielding is not an efficient solution to GCR chronic exposure, and the community consensus [25] is that pharmacological countermeasures, mission timing (solar maximum launch), or advanced shielding materials (polyethylene, liquid hydrogen, water walls) may offer better mass-efficiency trade-offs. Polyethylene at 16 g cm^{-2} delivers $\sim 26\%$ lower dose than aluminum at the same areal density, primarily because its hydrogen content fragments HZE ions more effectively. Detailed material optimization was beyond the scope of this work.

In contrast, the SEP analysis (Figure 3) shows that shielding is *highly effective* against the softer proton spectra of SEP events: the October 2003 Halloween event drops from $\sim 35\times$ the BFO limit at zero shielding to below the limit at $\sim 25 \text{ g cm}^{-2}$ Al. This asymmetry (GCR poorly shielded, SEP well shielded) strongly suggests that the most mass-efficient architectural choice is a compact, heavily shielded storm shelter for SEP events rather than whole-vehicle HZE shielding. The August 1972 event, if encountered without a shelter, would be potentially lethal; its spectrum requires $> 30 \text{ g cm}^{-2}$ for sub-limit protection. These numbers are consistent with prior analyses [21] and are reproduced here with an open, reproducible implementation.

4.3 REID Interpretation and Limitations

The median REID of 5.0% (male) and 5.4% (female) for a 35-year-old crew member on a 259-day transit at 16 g cm^{-2} aluminum exceeds the historical NASA 3% career limit [5]. However, several caveats are required.

First, the uncertainty bands are wide: the p5–p95 range spans approximately 1.3%–16.9%

(male), a factor of $\sim 13\times$ spread. A REID estimate of 5.0% median with this spread should not be read as a precise prediction; it should be read as: *the mission dose is likely to produce a meaningful cancer mortality risk, with the exact value uncertain by an order of magnitude under reasonable assumptions about the radiobiology.*

Second, the Cucinotta (2013) model was NASA’s primary risk model at the time of this work, but NASA has since updated its career limit framework to be age- and sex-dependent rather than a fixed 3% threshold [26]. Under the updated framework, mission go/no-go determinations require age- and sex-specific analysis; the pre-2023 3% threshold shown in Figure 4 is a historical reference only and should not be used to draw operational conclusions from the median REID values reported here. We do not implement the updated framework because it postdates the primary literature we built against; this is a meaningful limitation for any operational application of these results, and readers should treat the limit comparison as indicative rather than prescriptive.

Third, and most importantly, the dominant uncertainty in REID is biological, not physical. The variance decomposition from the LHS ensemble consistently shows that the DDREF (dose and dose-rate effectiveness factor) and the ERR/EAR scale parameters together account for $> 70\%$ of the output variance in REID, while physics parameters (solar modulation, LIS normalization, cross-sections) contribute $< 30\%$ combined. This result is consistent with Cucinotta et al. [5] and with the broader literature on space radiation risk [25]: reducing measurement uncertainty in GCR spectra (through improved LIS models or better solar modulation databases) will have limited impact on REID uncertainty without matching progress in our understanding of high-LET human radiobiology. There is currently no direct in vivo human data for the iron-ion LET range ($\sim 400 \text{ keV}/\mu\text{m}$) that contributes meaningfully to GCR dose equivalent in unshielded or lightly shielded geometries. The extrapolation from mouse tumor data and atomic-bomb survivor epidemiology to the HZE component of the GCR field is the most uncertain aspect of the entire risk calculation, and no amount of improved transport modeling can address it.

The sensitivity decomposition reveals an additional, interpretively important asymmetry. The solar modulation scale parameter (`phi_scale`) accounts for 72% of absorbed dose variance but only 3.5% of REID variance (Table 9). This means that uncertainty in ϕ — which is

substantial (± 50 – 100 MV from neutron monitor inter-station offsets, Section 2.2) — strongly influences how much physical dose a crew member receives but has almost no leverage on the probability that they die from it. Once dose is converted to REID through the organ EAR/ERR functions and life tables, the radiobiological scale parameters dominate by a wide margin and wash out the physics spread. Investment in better solar modulation databases or improved GCR spectral measurements will therefore reduce uncertainty in mission dose planning but will not narrow cancer risk estimates until the high-LET radiobiology is better constrained.

4.4 Per-Species LET and the Bridge to Proton Therapy

The LET decomposition in Figure 2 illustrates the physical mechanism behind the GCR effective quality factor. At $\phi = 481$ MV with no shielding, carbon ions (though contributing only $\sim 1\%$ of the GCR particle fluence) account for approximately 58.6% of the total absorbed dose. This arises because carbon LET (~ 25 keV/ μm at transit energies) is high enough to deposit significant dose per unit fluence, but the carbon flux is still large enough (relative to heavier ions like Fe and Si) to dominate the dose integral. The result is a field $\bar{L}_D \approx 18.7$ keV/ μm , yielding $Q_{\text{eff}} = 2.60$ via the ICRP-60 quality factor curve, consistent with the MSL/RAD measured value of 2.62 ± 0.14 [6].

After 16 g cm^{-2} of aluminum, nuclear fragmentation preferentially removes the heavy ions (longer mean free paths, higher cross-sections), and the carbon dose fraction drops to 31.2% while the proton fraction rises to 49.4%. The field $\bar{L}_D$ falls to 4.1 keV/ μm .

Proton therapy beams in the plateau region have $\bar{L}_D \sim 0.5$ – 2 keV/ μm , giving RBE ≈ 1.05 – 1.10 under the Wedenberg and McNamara models. Near the Bragg peak ($\bar{L}_D \sim 10$ – 25 keV/ μm), proton RBE rises to 1.4–1.8 (Figure 5) — the same LET range as the dominant GCR HZE dose regime at moderate shielding. The LET-RBE relationship that determines $Q(L)$ for astronaut risk is the same one that governs RBE uncertainty in proton therapy Bragg peak delivery, particularly for CNS and late-responding tissues where $\alpha/\beta \approx 3$ Gy [12].

The transport physics for a clinical proton beam (monoenergetic, narrow field, tissue-equivalent phantom) and for GCR (isotropic, broad spectrum, shielded spacecraft) are quite different, and the pipeline does not bridge that physical gap. What it does show is that the

same LET-dependent RBE formalism can be evaluated consistently for both scenarios from a single codebase, enabling direct numerical comparison of quality factors across the two fields — a starting point for more unified treatments of high-LET radiation biology in space and clinical contexts.

4.5 Limitations

The primary limitations of the pipeline are collected here; several have been noted briefly in the methods.

4.5.1 Transport Model Approximations

The CSDA + exponential attenuation transport model does not account for:

- **Nuclear fragmentation secondaries.** When a heavy ion (e.g., Fe) undergoes a nuclear interaction with an Al nucleus, it fragments into lighter ions (C, O, N, He, p) that continue propagating with lower LET. The current model treats the interaction as simple fluence removal (exponential attenuation), not as a source of lighter-ion secondaries. This leads to underestimation of secondary proton and helium flux at depth, and overestimation of iron fluence attenuation. The effect is largest at $> 30 \text{ g cm}^{-2}$, where the pipeline’s shielding curve may underestimate dose reduction relative to a full fragmentation model. Comparisons between simplified transport and full fragmentation codes in the literature [4, 11] suggest that omitting secondary production overestimates the effective quality factor at 16 g cm^{-2} by approximately 15–25%, because fragmentation progressively converts high-LET primary ions into lower-LET secondaries. The resulting overestimation of REID is likely of similar fractional magnitude — well within the $13\times$ p5–p95 uncertainty band, but not negligible relative to the median.
- **Delta-ray production.** Energetic secondary electrons (delta rays) from ion tracks are not tracked. For dose and dose equivalent at the macroscopic level, this is a small correction; for microscopic track structure calculations relevant to DNA damage models, it would matter.

- **Geometry simplification.** The spacecraft is modeled as a uniform aluminum slab of fixed areal density. Real crew vehicle geometries are inhomogeneous: walls vary in thickness and material across directions, equipment provides additional shielding in some directions, and crew members are not at the geometric center. Ray-tracing geometry models (as in OLTARIS or Geant4) would be needed to address this.
- **Stopping power uncertainties.** The NIST PSTAR/ASTAR stopping powers for aluminum and tissue are accurate to $\sim 1\text{--}2\%$ above 1 MeV/n, but less certain at lower energies where shell corrections become important. This introduces a small systematic uncertainty in CSDA range calculations for low-energy ions near the end of their range.

4.5.2 Spectrum and Modulation

- **Force-field approximation.** The Gleeson–Axford force-field model ignores charge-sign-dependent drift and diffusion anisotropy. During solar minimum, these effects can modify the GCR spectrum below ~ 500 MeV/n at the 10–20% level [27], which is within the LIS normalization uncertainty we propagate but is not explicitly modeled.
- **Missing species: Magnesium.** The pipeline currently models H, He, C, O, Si, and Fe. Magnesium ($Z=12$, $A=24$) is a minor GCR contributor with an LIS spectral index and abundance tabulated in Boschini et al. [3] but not yet implemented. Its contribution to total absorbed dose is estimated at $< 3\%$, based on the Boschini (2020) Mg LIS normalization scaled by its Z^2 -weighted stopping power relative to the six modeled species at ~ 500 MeV/n.
- **Anomalous cosmic rays.** At energies below ~ 100 MeV/n, anomalous cosmic rays (ACRs, singly-ionized interstellar neutrals accelerated at the solar wind termination shock) can contribute meaningfully during solar minimum. ACRs are not modeled here.

4.5.3 Biological and Risk Model

- **NASA 2023 risk framework.** The pipeline implements the Cucinotta (2013) risk model. NASA’s 2023 update introduces age- and sex-dependent career limits and

revised tissue weighting factors. Quantitative comparison with the updated framework is left to future work.

- **High-LET extrapolation uncertainty.** The EAR and ERR functions in the REID model were fit primarily to low-LET (photon) epidemiological data (atomic-bomb survivors) with high-LET scaling from rodent experiments. The HZE component of GCR has no direct human epidemiological data to constrain risk models. The DDREF, which bridges the low-dose-rate space environment and the acute exposures in the underlying datasets, is one of the most debated parameters in space radiation risk assessment [20].
- **Non-cancer risks.** The REID framework models cancer mortality only. Central nervous system (CNS) effects from HZE ions, cataracts, cardiovascular risk, and heritable genetic effects are not quantified here. CNS effects in particular have received increased attention following rodent studies showing cognitive impairment at doses well below the cancer threshold [28].
- **Sex and age scope.** Results are shown for a 35-year-old male and female. The age dependence of REID is significant: a 25-year-old female has approximately 1.5–2× higher REID than a 45-year-old male for the same exposure, because she has more remaining life years in which cancer can develop and because female breast and ovarian tissues have high tissue weighting factors. The pipeline supports arbitrary age and sex inputs; broader demographic analysis is left for future work.

4.5.4 TOPAS Benchmark Scope

The TOPAS benchmark in this paper tests whether the pipeline computes the same RBE values as OpenTOPAS-RBE given the same LETd input. It does not test the physical transport. The LETd profiles used as input were generated from the TOPAS-nBio V79 reference dataset for V79, and from the V79 LETd spectrum propagated through the clinical cell line parameters for prostate and H&N. The small mean absolute errors reported ($< 10^{-3}$) therefore reflect numerical accuracy of the implementation, not independent physical validation. A full benchmark comparing pipeline-predicted versus TOPAS-Monte-Carlo

depth-dose and LETd profiles in the same phantom geometry is left for future work when a TOPAS binary is available in the CI environment.

4.6 Comparison with Existing Tools

Table 8 summarizes the principal differences between this pipeline and the major existing GCR dosimetry tools. The pipeline is not intended to replace these tools for high-fidelity mission analysis; it is intended as a transparent, reproducible starting point for researchers who want to understand the physics or run uncertainty analyses without access to proprietary codes.

Table 8: Comparison of GCR dosimetry tools. $\checkmark$ = present, $\triangle$ = partial, $\times$ = absent.

Feature	Ours	HZETRN	OLTARIS	Geant4	TOPAS
Open / pip-install	$\checkmark$	$\times$	$\times$	$\checkmark$	$\checkmark$
Fragmentation	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Organ REID	$\checkmark$	$\triangle$	$\checkmark$	$\times$	$\times$
SEP library	$\checkmark$	$\triangle$	$\checkmark$	$\times$	$\times$
LHS uncertainty	$\checkmark$	$\times$	$\times$	$\times$	$\times$
Clinical RBE	$\checkmark$	$\times$	$\times$	$\triangle$	$\checkmark$
CI test suite	$\checkmark$	$\times$	$\times$	$\times$	$\times$

4.7 Reproducibility

All results in this paper can be reproduced by cloning the repository and running:

```

pip install -e .[dev]
python scripts/validate_pipeline.py
python scripts/validate_extended.py
pytest

```

The five publication figures are generated by the scripts in `scripts/plot_*.py`. The full N=500 uncertainty ensemble requires approximately 90 minutes on a single CPU core; a reduced N=20 version runs in under 2 minutes and is used in the automated CI pipeline.

Input data (Usoskin ϕ database, NIST stopping powers, HZETRN neutron table) are included in the repository under `data/`. No proprietary datasets are required.

5 Conclusion

We have presented GCR-DOSIMETRY-PIPELINE, an open-source Python package installable via pip that integrates GCR spectrum generation, CSDA slab transport, organ-specific REID, SEP acute dosimetry, Latin Hypercube uncertainty propagation, and proton therapy RBE benchmarking into a single reproducible workflow. The pipeline is calibrated to MSL/RAD cruise-phase dosimetry and shows consistency with available HZETRN benchmarks within their stated uncertainties.

The central quantitative results are:

- A 259-day Earth–Mars transit at 16 g cm^{-2} aluminum delivers a median REID of $\sim 5\%$, with p5/p95 bounds of 1.3% – 16.9% , exceeding the historical NASA 3% limit; the wide uncertainty interval is dominated by DDREF and excess risk scaling, not by GCR transport physics.
- The August 1972 SEP event would require $> 30 \text{ g cm}^{-2}$ aluminum for sub-limit BFO protection; at 16 g cm^{-2} (ISS/Orion class), the dose is estimated at $\sim 4\times$ the limit.
- Carbon ions account for $\sim 59\%$ of unshielded GCR absorbed dose at $\bar{L}_D = 18.7 \text{ keV } \mu\text{m}^{-1}$; 16 g cm^{-2} aluminum reduces this to 31% and drops the field $\bar{L}_D$ to $4.1 \text{ keV } \mu\text{m}^{-1}$, within the LET range of clinical proton therapy Bragg peaks.
- The Wedenberg and McNamara RBE models, computed from the same $\bar{L}_D$ infrastructure as the GCR quality factor, agree with OpenTOPAS-RBE to within $\text{MAE} < 10^{-3}$ for V79 cells (self-consistency check for clinical cell lines).

The pipeline has significant limitations relative to existing high-fidelity codes (most importantly, no nuclear fragmentation tracking) and should be treated as a transparent starting point with explicit uncertainty quantification, not a replacement for HZETRN or Geant4. Future work will focus on adding fragmentation cross-sections from the Wilson

et al. database, implementing the NASA 2023 risk framework, and running full TOPAS Monte Carlo simulations using the benchmark geometries provided in the repository. All source code, 103 automated tests, validation datasets, and TOPAS input files are available at <https://github.com/aryanhshah8/gcr-dosimetry-pipeline>.

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A LHS Sensitivity Decomposition ($N = 500$)

Table 9 lists the Spearman ρ^2 sensitivity indices for each of the eight LHS parameters with respect to REID, absorbed dose D , and dose equivalent H , from the $N = 500$ ensemble at 16 g cm⁻² aluminum, 259-day transit, 35-year-old male.

Table 9: Normalized Spearman ρ^2 sensitivity indices. Values sum to 1 within each column by construction. Radiobiological parameters dominate REID variance; physics parameters dominate D and H variance.

Parameter	REID	D	H
phi_scale	0.035	0.720	0.050
LIS_norm	0.013	0.266	0.027
cross_sec	0.007	0.007	0.002
neutron_H	0.033	0.002	0.026
Q_factor	0.764	<0.001	0.894
DDREF	0.067	<0.001	<0.001
ERR_scale	0.020	0.001	<0.001
EAR_scale	0.061	0.004	0.001

$N = 500$ samples; 35-year-old male, 259-day transit, 16 g cm^{-2} Al.

CRedit Author Contribution Statement

Aryan Hemendra Shah: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of Competing Interests

The author declares no competing interests.

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Data Availability Statement

All source code, input datasets, validation scripts, and TOPAS benchmark geometry files are freely available at <https://github.com/aryanhshah8/gcr-dosimetry-pipeline>. No proprietary datasets were used; all input data (Usoskin solar modulation potential database, NIST stopping powers, HZETRN neutron benchmark table) are included in the repository under `data/`. Results can be fully reproduced without any proprietary software.

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